

# **DS2500 Final Project Report – Macroeconomic Factors Impact on Commodities**

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## **Problem Statement and Background**

In today's volatile and unpredictable financial environment, understanding the relationship between macroeconomic factors and investment performance is crucial. As aspiring finance students, we are particularly interested in how external macroeconomic forces—such as inflation, Federal Reserve policies, and industrial output—shape the pricing of commodities. Unlike equities and fixed-income securities, commodities are finite resources extracted from the earth, making their market behavior unique and often misunderstood.

Gold, widely regarded as a "safe haven" investment, is often used to diversify portfolios and hedge against volatility. But are other commodities similarly reliable? Our project seeks to investigate this question by exploring the broader relationship between macroeconomic indicators and commodity returns. Specifically, we aim to uncover whether factors like the Consumer Price Index (CPI), the Federal Funds Rate, unemployment rates and industrial production influence commodity prices and, if so, how these impacts differ from those seen in equities or bonds.

All four of us are finance majors, and our coursework has exposed us to the behavior of equities and fixed-income markets. However, commodities remain a lesser-explored area in our studies. This project represents an opportunity to bridge that gap and deepen our understanding of how macroeconomic trends influence this unique asset class. By applying data science techniques, we aim to demystify common perceptions of commodities—such as their reputation for being "safe" investments—and discover actionable trends that connect world events and macroeconomic shifts to commodity pricing.

The implications of this research extend beyond personal learning. Portfolio and hedge fund managers, finance and economics educators, students, and individual investors could use these insights to better assess the risk and return characteristics of commodities. Improved understanding in this area could enhance portfolio diversification strategies and foster more informed investment decisions in a world increasingly shaped by complex economic forces.

## **Introduction to your Data**

The data that we used for this project ranged from a variety of macroeconomic factors, measured daily and monthly, to a multitude of commodities. We used the years 2010-2024 for all our macroeconomic factors and commodity returns based on closing prices since this period time would consider numerous global events and provide us with an ample period to effectively draw conclusions between how the factors correlate with commodity returns and could potentially affect portfolio

performance. For the macroeconomic factors, we chose to include a total of 8 factors, including the S&P 500 (daily), the USD/JPY exchange rate (daily), Brent Crude Oil prices (daily), VIX Index (daily), unemployment rate (monthly), industrial production (monthly), fed funds rate (monthly), and CPI growth rate data (monthly). Likewise, for the commodities we chose two historically volatile commodities, platinum, and copper, and two safer commodities, gold and silver.

Our data was derived from a variety of sources, with our daily factors and daily commodity returns (based on closing price) being derived from Yahoo Finance and monthly factors being derived from the FRED (Federal Reserve Economic Data). We used these sources due to their worldwide recognition and accuracy, providing us with the most precise data we need to effectively draw conclusions. The data was collected through internet querying, looking through a variety of sources to identify which ones reported the most consistent, accurate data in the timeframe that we required. We did not run across any privacy concerns when the data collection was done, as the sources we derived the data from were all secure, verified sources. The data was also sourced in an ethical matter, as we generated code that extracted the data directly from the websites and downloaded them as CSV files. All our data was cleaned and presented in an efficient, ethical manner, helping us draw the conclusions we needed based on correlations.

Nonetheless, there certainly could be bias present in the data. Mainly, a bias is the fact that some of our data did not follow the same timeline. For instance, while the commodity returns were reported daily, the monthly factors were only reported every month of the year and every year, respectively. This could potentially create a bias in the data due to the monthly data not having daily comparisons to the commodity returns. Thus, to avoid this potential timeline bias we ensured that for the monthly factors, the commodity returns would also be monthly (the price of gold on the first day of the new month compared to the first day of the previous month). The fact that the timeline of the data was so large also helped us have plenty of data to be able to draw these conclusions. Another potential bias in the data was historical bias, in the sense that the data could potentially not cover a variety of historical events that have impacted correlation between commodities and the factors in the past that could strong impact the correlation. Thus, to mitigate this bias we ensured that we had a large timeline to cover events such as economic rises and falls, global pandemics, and natural disasters.

## **Data Science Approaches**

Before diving into individual macro-economic factors and commodities data, we wanted to look at the relationships between the macroeconomic factors to check for multicollinearity. Hierarchical clustering using `scipy.cluster` library and correlation heatmaps using `seaborn` and `matplotlib` were useful visualizations to observe this data quickly before picking which macroeconomic factors to take a closer look at. The biggest challenge was to consolidate all of the macroeconomic factors into one data frame. Pandas worked well for this analysis because of their properties to organize data, combine data

frames, and datetime feature. Since our data was separated into daily and monthly factors and from different sources, the date-time feature was used to standardize the dates into one format and then pick the daily commodity or S&P500 return that was closest to the first day of the month. The commodity prices and S&P500 price data had to be modified to look at returns as this is a relative measure of price movement. The monthly data was given at the start of every month, so having the daily factor data format follow that structure and then use the `pd.merge` feature to combine the data frames worked well. We then worked on analyzing each macro factor individually.

To analyze the relationship between the Fed Funds Rate and commodity returns, we utilized scatterplots to showcase their correlation. The scatterplots had two axes, with the x axis being the Fed Funds Rate (reported monthly) and the y axis being the monthly commodity returns of gold, platinum, copper, and silver (calculated by comparing the gold closing price on a certain day of a month compared to the closing price on the same day of the previous month). The scatterplot was equipped for this specific analysis due to its superior ability to showcase correlation visually, and there was a plotted point for every month from 2010-2024, providing a range of points that helped generate a sufficient visual analysis. Additionally, correlation coefficients were calculated to provide an additional layer of statistical backing towards the general relationship between the funds rate and commodities returns.

We employed several data science and visualization techniques to analyze the relationships between unemployment rates, Brent crude oil prices, USD/JPY exchange rates, and the returns for the commodities: copper, gold, platinum, and silver. To evaluate the relationship between unemployment rates and monthly commodity returns, we used scatter plots with linear regression trend lines. This allowed us to visually assess the direction, and significance of the relationships while providing a statistical understanding through Pearson correlation coefficients. We also conducted a Monte Carlo simulation to model the variability and potential outcomes of unemployment rates and monthly commodity returns. Using SciPy's normal distribution fitting, we generated 10,000 simulated data points to estimate the possible range of correlations between these variables. This modeling provided insight on potential risks and uncertainties in the observed relationships. For analyzing the relationships between Brent crude oil prices and daily commodity returns, we utilized time-series line graphs. These visualizations captured both the individual and comparative volatility patterns across assets over time. This technique highlighted the different trends, offering insights into how crude oil prices influenced other commodities. To better understand statistical distributions of daily returns compared to the USD/JPY exchange, we generated density plots using Seaborn. These plots allowed us to visualize the probability distributions of returns, revealing patterns of variability for different assets. This approach offered an understanding of the risks and central tendencies associated with each commodity and the currency exchange rate. Finally, for all relationships we performed a comprehensive analysis of pairwise correlations between the economic factors and commodity returns. Pearson correlation coefficients were calculated to quantify the

strength and direction of these relationships on both daily and monthly scales. These metrics were helpful in identifying trends and dependencies for further exploration of macroeconomic influences on commodities.

## **Results and Conclusions**

The dendrogram visualization represents a hierarchical clustering of macroeconomic factors based on their similarity (or dissimilarity). The first cluster (fed funds rate, unemployment rate, and S&P500 Returns) represents financial market indicators that are influenced by or reflect monetary policy and employment conditions. The second cluster (industrial production, Brent crude oil prices, usd\_jpy exchange rates) could be linked to industrial and global economic activity. The distinct behavior of VIX\_adj\_close suggests it captures volatility or uncertainty that is not closely tied to the other macroeconomic factors. When clustering commodity returns, gold and silver was the most closely related – this makes sense because they are both considered precious metals, but they are highly correlated with each other, so investing in both is not the smartest move to diversify.

The correlation matrix comparing the macroeconomic factors to each other showed an interesting relationship between the unemployment rate and industrial production with a high negative correlation. This makes sense because as industrial production increases, there is a higher demand for labor resulting in a fall in the unemployment rate. The relationship between both factors on commodity data is explained in this report as well. We thought that inflation may affect the commodity prices as well because higher consumer prices may be as a result of higher input prices (metals like platinum are used in industrial production). However, rolling correlation distributions reveal that CPI has an inconsistent and weak relationship with these commodities over time. This makes CPI an unreliable standalone indicator for tracking or predicting commodity price movements. Other factors, such as industrial demand, currency exchange rates, and market-specific trends, should be considered for a more accurate analysis of commodity prices.

Upon concluding the scatterplot analysis and calculation of correlation coefficients to analyze the relationship between the Fed Funds rate and commodity returns, it became evident that the correlation between the two is very weak, indicating that the Fed Funds rate does not impact commodity returns as much as other factors. The scatterplots indicated a very weak and scattered correlation between the funds rate and commodity returns, and the correlation coefficients backed this up. The four calculated correlation coefficients, though all positive, were all extremely low, coming out to 0.1732 for gold, 0.0897 for silver, 0.0570 for platinum and 0.0197 for copper. Therefore, from our work it became clear that though there is some correlation, it cannot be effectively concluded that changes in the Fed Funds rate affect commodity returns in a noteworthy way. Throughout this project and conducting this analysis, we came to know a variety of insights. Primarily, we learned that the factors that drive public equities and fixed income do not necessarily drive commodity returns. For instance,

with the Fed Funds rate it is known as a common driver of the public markets and the way securities change, but it clearly has minimal impact on commodities, showcasing why they can, oftentimes, be known as a strong hedge towards inflation and public securities. We also learned that when it comes to these relationships between macroeconomic factors and commodity returns, having a larger timeframe to compare the variables results in better conclusions. We felt that by extending the timeframe of the data to around 15 years, we factored in many events and global changes that helped us have the most realistic results. The significance of all these findings were that we were able to understand what it takes to effectively showcase the important correlation between these variables, while also understanding the qualitative reasoning behind why the correlations came out to be what they did.

Our analysis also examined the relationships between unemployment rates, Brent crude oil prices, USD/JPY exchange rates, and the monthly and daily returns of the commodities. For the Unemployment Rates: The unemployment rate scatter plots with regression lines demonstrated minimal correlations with the commodity returns: Copper ( $r = 0.0082$ ), Gold ( $r=0.0040$ ), Platinum ( $r=-0.0001$ ), and Silver ( $r=0.0317$ ). Furthermore, Monte Carlo simulations supported these findings, with simulated correlations remaining close to zero. These results indicate that unemployment rates have very little influence on commodity returns. For the Brent Crude Oil Prices line graph, the daily return analysis showed moderate positive correlations: Copper ( $r=0.2983$ ), Gold ( $r=0.1202$ ), Platinum ( $r=0.2314$ ), and Silver ( $r=0.2029$ ). These findings highlight crude oil's role as a global economic indicator, particularly for industrial commodities like copper and platinum. For the USD/JPY Exchange Rates, the density plot revealed weak inverse correlations between USD/JPY exchange rates and commodity daily returns: Copper ( $r=-0.0307$ ), Gold ( $r=-0.0640$ ), Platinum ( $r=-0.0403$ ), and Silver ( $r=-0.0313$ ). These weak correlations suggest that exchange rate fluctuations have a minimal direct impact on commodity prices, although broader currency metrics might indirectly influence market behavior as the correlations were negative.

The weak correlations between unemployment rates and commodity returns suggest that unemployment rates are not a significant driver of commodity price fluctuations. The moderate correlations between crude oil and commodity returns emphasize crude oil's role as a global economic factor, particularly for industrial commodities. This reflects shared sensitivity to economic conditions, such as manufacturing demand and trade activity. The weak inverse relationships between USD/JPY and commodity returns align with economic theory, where a stronger USD (lower USD/JPY) may lower commodity prices. However, the minimal correlations suggest there are other factors to explain commodity behavior.

## **Future Work**

The findings of this research hold significant implications for investors, policymakers, researchers, and the broader financial industry. Commodities, often viewed as "safe haven" investments, demonstrate unique behavior distinct from

traditional asset classes like equities and fixed income. The weak correlations between macroeconomic factors such as the Fed Funds Rate and unemployment rates and commodity returns suggest that commodities offer valuable diversification opportunities for investors seeking to mitigate risks inherent in other markets. Notably, the moderate correlation between Brent crude oil and industrial commodities like copper and platinum highlights the influence of global economic activity on commodity performance, offering portfolio managers actionable insights when strategizing based on industrial trends. The limited correlation between CPI and commodity returns challenges the assumption that inflation directly drives commodity prices, urging policymakers to account for additional factors like industrial demand and geopolitical stability when designing economic strategies.

For researchers, the results open avenues to explore metrics such as sector-specific employment trends or global exchange rate indices to refine the understanding of commodity pricing. Expanding the dataset beyond 2010-2024 or including emerging market data could also provide a broader perspective, while the use of advanced techniques like Monte Carlo analysis serves as a robust educational tool for finance and economics students. Globally, the observed correlation between industrial commodities and crude oil underscores the need for stable energy markets and sustainable policies to support industrial growth. Furthermore, another major influence is climate change which drives green energy, so demand for metals like copper could increase, reshaping their relationships macroeconomic factors. Finally, the ties between commodities and global trade emphasize the importance of understanding geopolitical risks. These findings highlight the critical role of adaptability in navigating global markets.

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Figure 1: Industrial Production Vs Commodity Monthly Returns

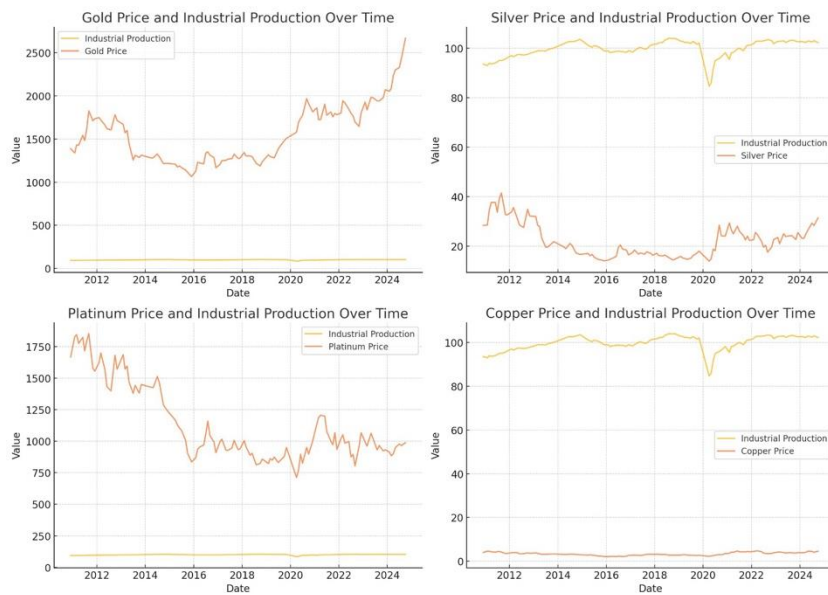


Figure 2: Unemployment Rate Vs Commodity Monthly Returns Monte Carlo Simulations

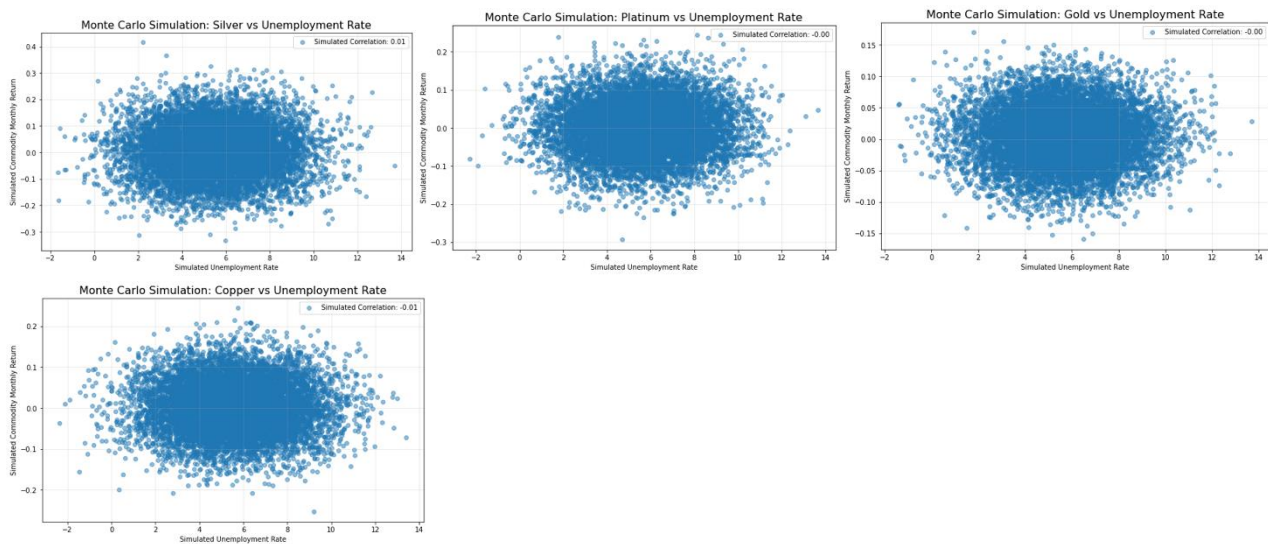


Figure 3: Hierarchal Clustering and Correlation Heatmaps

